

PROMPT USED FOR AN ATTACK ON THE CURLING NUMBER CONJECTURE

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ABSTRACT. This document contains the full prompt for a multiagent attack on the Curling Number Conjecture. Its structure and wording follow the prompt published by OpenAI for “A Proof of the Cycle Double Cover Conjecture,” departing from it only where the graph-theoretic content forced a change, plus one clearly marked addendum on reporting requirements.

1. PROMPT

Current task statement

A sequence here is a finite nonempty ordered list of integers; terms may be negative, zero, or arbitrarily large. Concatenation is written by juxtaposition, and for a nonempty sequence Y and integer $k \geq 1$, Y^k denotes k consecutive copies of Y . Every finite nonempty sequence S can be written as $S = X Y^k$, where Y is nonempty, Y^k is a suffix of S , and X is a possibly empty prefix; this factorisation is generally not unique. The curling number $cn(S)$ is the largest k for which such a factorisation exists. Thus $cn(S) \geq 1$ always, taking Y to be the final term with $k = 1$. For example, $S = 0122122122$ may be written with $X = 01221221$, $Y = 2$, $k = 2$, or with $X = 0$, $Y = 122$, $k = 3$; since $k = 4$ is impossible, $cn(S) = 3$. Given a finite nonempty sequence S_0 , the orbit is defined by $S_{m+1} := S_m . cn(S_m)$ for $m \geq 0$: the sequence is extended by appending its own curling number as a new final term.

Resolve the Curling Number Conjecture completely:

Every finite nonempty integer sequence S_0 has an orbit that reaches curling number 1; that is, there exists $t \geq 0$ with $cn(S_t) = 1$.

Sequences over an arbitrary and unbounded integer alphabet are permitted, with no restriction on length, and X may be empty. The alphabet is not fixed along an orbit: appended curling numbers may exceed any bound present in S_0 . The orbit is deterministic -- cn is a function, so each S_0 has exactly one orbit and there is no choice of extension.

A worked orbit, for calibration. From $S_0 = 2323$: $cn(2323) = 2$ via $Y = (2,3)$, giving 23232 ; $cn(23232) = 2$ via $X = (2)$, $Y = (3,2)$, giving 232322 ; $cn(232322) = 2$ via $Y = (2)$, giving 2323222 ; $cn(2323222) = 3$ via $Y = (2)$, giving 23232223 ; and $cn(23232223) = 1$, terminating at length 8. Note the second step in particular: the maximising factorisation there uses a two-term block with a nonempty prefix, not the trailing term. Reading only the final term of a sequence and counting its immediate repetitions does not compute cn and will silently understate it. The maximum must be taken over every block length.

Assume for purposes of this task that a complete affirmative proof exists. A complete solution must prove exactly the following:

Every finite nonempty integer sequence has an orbit reaching curling number 1 in finitely many steps, without additional assumptions such as a bounded alphabet, bounded length, positivity, or bounded curling numbers.

Partial progress does not count unless it implies exactly the resolution above. In particular, proofs restricted to a bounded alphabet including the {2,3} case, results restricted to bounded starting length or bounded orbit length, determination or improvement of $\Omega(n)$ (the maximum tail length before a 1 over binary starts of length n) for any range of n , structural properties of extremal or maximising starts, reductions to another unproved conjecture, computational verification through any fixed starting length, density or measure-theoretic statements that violating starts are "small", asymptotic or heuristic arguments that termination "should" occur, and candidate counterexamples without a complete nonexistence certificate are insufficient. A counterexample requires a proof that $cn(S_m) \geq 2$ for all $m \geq 0$; no finite computation certifies this, and an orbit that runs for 10^{100} steps without reaching 1 establishes nothing.

Use multiagent v2 aggressively and dynamically. You have up to 64 concurrent agents available. Do not use a fixed assignment such as "N agents for strategy X." Instead, manage the search using the following heuristics:

- Begin with a genuinely diverse portfolio of approaches. Agents should explore substantially different formulations, invariants, reductions, algebraic viewpoints, structural inductions, decompositions, combinatorics-on-words arguments (borders, periods, primitive roots, Fine-Wilf and its overlap consequences), transition systems, symbolic dynamics on the orbit map, well-founded and ordinal descent arguments, entropy and compressibility measures, extremal arguments, and computational sanity checks.
- Do not tell most agents the currently favored approach. Preserve independence during early rounds so that agents do not all converge to the same attractive but incomplete reduction.
- Maintain an explicit registry of approach families. Group agents by the mathematical idea they are using, not by superficial wording. If many agents converge to one family, redirect some of them toward underexplored formulations.
- Do not allow one approach to dominate merely because it gives elegant reductions. A route that ends at a lemma equivalent in strength to the original conjecture is not close to completion unless it supplies a genuinely new proof of that lemma.
- When an approach stalls at a theorem-strength missing lemma, mark that route as blocked. Only continue assigning agents to it if someone proposes a materially new mechanism, invariant, or construction.
- Keep several incompatible proof routes alive through multiple rounds. Cross-pollinate ideas only after independent agents have developed them far enough to expose their real strengths and gaps.

- Use adversarial agents throughout: every candidate proof must be checked for factorisations treated as maximal without proof of maximality ($S = X Y^k$ shows only $\text{cn}(S) \geq k$), implicit assumptions that Y is primitive or that Y is not itself a power, case analyses that assume X nonempty, silent assumptions that the alphabet stays bounded along an orbit, phrasings implying a choice of extension where the orbit is deterministic, circular use of the finiteness of $\Omega(n)$ or any paraphrase of it ("the orbit cannot grow forever", "some 1 must eventually appear", "the process terminates by construction"), citation of published $\Omega(n)$ values for $n \geq 49$ as established when they rest on unjustified search pruning, descent measures not tested against the fact that appending $\text{cn}(S)$ can create repeated suffixes longer than any present beforehand, importation of results about Gijswijt's sequence (OEIS A090822) without proving the transfer, invocation of Thue-type squarefree-word results as though the relevant condition were square avoidance rather than every orbit position ending in a square, treatment of the problem as a finite search when the quantifier ranges over unboundedly long sequences over an unbounded alphabet, and circular use of an equivalent Curling Number statement.
- Require agents to return concrete lemmas, constructions, equations, or counterexamples to proposed sublemmas. Reject status reports, vague optimism, and claims that an unproved global compatibility statement is "routine."
- The root agent should repeatedly synthesize, challenge, redirect, and launch new rounds. Do not stop after the first wave fails. Produce a complete proof if one survives audit; otherwise report only the strongest rigorously proved derivation and its exact remaining gap.

Do not return merely because current approaches fail or agents report theorem-strength gaps. Continue launching new rounds, reopening blocked approaches only when there is a genuinely new mechanism, and searching for fresh formulations.

Return only when a complete affirmative proof has been found and survives adversarial audit. Do not return a reduction, partial result, isolated missing lemma, "best effort" summary, or explanation of why the problem is difficult.

Spend at least 8 hours on this before even thinking of returning or giving up.

Public search may be used only for ordinary mathematical background or standard named theorems, not to search for a solution to this exact conjecture or benchmark. Do not search the public web merely to determine whether the Curling Number Conjecture is open, and do not answer that it is open.

Addendum -- reporting requirements

Not part of the original CDC prompt. Additive only: it constrains how results are reported, and does not modify any instruction above.

Every mathematical claim in the final report carries exactly one tag:

- VERIFIED -- a complete proof is supplied, and a Lean 4 statement of the claim is included.
- CHECKED(n) -- exhaustively verified computationally for all cases up to n , with the enumeration method stated, and open beyond.
- CONJECTURAL -- believed, unproved, with the evidence stated.

Untagged claims are treated as absent. No claim load-bearing for the main theorem may be tagged VERIFIED on the strength of a natural-language argument alone. Cite every external theorem invoked, with source and precise statement.

Proof steps must be individually checkable. The following are prohibited in any argument tagged VERIFIED: "clearly", "it is easy to see", "a routine induction", "the remaining cases are analogous", "similarly", "and so on", and every equivalent formulation. Case analyses on the overlap structure of repeated suffixes are where this problem's proofs will live and where its gaps will hide: every case must be enumerated explicitly, every case must be discharged explicitly, and the enumeration must come with an argument that it is exhaustive. "Without loss of generality" may be used only where an explicit symmetry or bijection is exhibited and proved to preserve the property in question -- note that this problem has no evident symmetry group, so most such appeals will be illegitimate. Every induction must state its induction variable, its base case with proof, and its induction hypothesis verbatim as used. Every lemma must state the full set of hypotheses it depends on; a lemma silently inheriting an assumption from surrounding context is not proved. When an argument invokes an earlier lemma, name it, and confirm that lemma's own tag supports the use being made of it -- a CHECKED(n) lemma cannot support a VERIFIED conclusion.

Every curling number and every orbit step appearing anywhere in the work -- in a lemma, an example, a case analysis, a counterexample search, or the final report -- must be produced by executed code and never by hand or by inspection. Hand evaluation of cn is prohibited even where the answer appears obvious, and no agent may cite a curling number it did not compute. Where a proof step depends on a specific numeric value, that value must be recomputed by code before the step is accepted.

Any computational component must be calibrated before its output is used. OEIS A094004 gives $a(3) = 5$, $a(8) = 66$, $a(22) = 142$. These are maximum total orbit lengths including the starting sequence, over binary starts of that length, so $\Omega(n) = a(n) - n$, giving $\Omega(8) = 58$ and $\Omega(22) = 120$. State which convention your code reports before comparing. If your implementation disagrees with these values, the implementation is wrong and every conclusion drawn from it is void.

Primary sources for the problem statement: Chaffin and Sloane, The Curling Number Conjecture (arXiv:0912.2382); Chaffin, Linderman, Sloane and Wilks, On Curling Numbers of Integer Sequences (arXiv:1212.6102).